

Intelligent Platform for Crowdsourcing (IPCrowd) Technology Preparations

Georgia Institute of Technology
Georgia Tech Research Institute (GTRI)

Carl Cox, Tiffany Huang, Alexandra Trani

carl.cox@gtri.gatech.edu
tiffany.huang@gtri.gatech.edu
alexandra.trani@gtri.gatech.edu



Purpose

Introduce/revisit some commonly used technology tools:

- Python
- Google Platforms – Google Colaboratory (Colab)
 - <https://colab.google/>
- Repository and Git version control
 - GT GitHub
 - <https://drupal.gatech.edu/handbook/github-georgia-tech>
 - <https://support.cc.gatech.edu/support-tools/faq/what-gt-github-enterprise>
 - GT GitLab
- IDE – VSCode
 - <https://code.visualstudio.com/>
- Figma
 - <https://www.figma.com/>
- Past Presentations
 - Students
 - Speakers
 - <https://gatech.instructure.com/courses/341956/files/>
- **NLP**
 - Introduction to NLP Presentation on Canvas
 - **Large Language Models (LLMs) – Hugging Face**
- **Flask**
 - **A simple app with LLM**

Large Language Models – Hugging Face

Overview:

- Introduce Hugging Face
- Hugging Face Transformer Models
- Installation and Setup
- Examples to Run
- Potential Models to Use
- References

[Hugging Face](#) is a repository containing large amount of open source Machine Learning models for fields including:

- computer vision
- natural language processing (NLP)
- audio
- multimodal
- reinforcement learning

In NLP Field:

- Support multiple languages
- Models include
 - Bidirectional Encoder Representations from Transformers (BERT) [1]
 - Denoising autoencoder for pretraining sequence-to-sequence model (BART) [2]
 - GPT-2 [3]
 - Lots more

							
Conversational	Fill-Mask	Question Answering	Sentence Similarity	Summarization	Table Question Answering	Text Classification	Text Generation
2,076 models	6,760 models	4,598 models	1,829 models	1,060 models	59 models	21,316 models	11,157 models

Hugging Face Transformer Models

- Transformers
 - Hugging Face NLP models trained on large data
- Text Classification Example:
 - <https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment>
 - <https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>
- Other Tasks:
 - Question answering
 - Zero-shot classification
 - Summarization
 - Text classification
- Quick Tour
 - <https://huggingface.co/docs/transformers/quicktour>
 - Choose either Pytorch, TensorFlow, Flax

Installation and Examples

Options (pick one):

1. Installation page:
<https://huggingface.co/docs/transformers/installation>
2. Follow the `README.md`

Example code on Canvas `transformers` code:

- Question answering task:
bart_question_answer.py
- Zero shot classification task:
bart_zero_shot_classify.py

bart_question_answer.py 2 X

bart_question_answer.py > ...

```

1 """
2 The script runs BART model on question answering task given a question and the answer in the text.
3 The model used could be found on:
4 https://huggingface.co/docs/transformers/model\_doc/bart#transformers.BartForQuestionAnswering.forward.example
5
6 """
7
8 from transformers import AutoTokenizer, BartForQuestionAnswering
9 import torch
10
11 tokenizer = AutoTokenizer.from_pretrained("valhalla/bart-large-finetuned-squadv1")
12 model = BartForQuestionAnswering.from_pretrained("valhalla/bart-large-finetuned-squadv1")
13 question, text = "Who was Jim Henson?", "Jim Henson was a nice puppet"
14 #question, text = "What's the largest animal in the world?", "Elephant, ancient, friendly and largest mammal in the world."
15 inputs = tokenizer(question, text, return_tensors="pt")
16
17 with torch.no_grad():
18     outputs = model(**inputs)
19
20 answer_start_index = outputs.start_logits.argmax()
21 answer_end_index = outputs.end_logits.argmax()
22 predict_answer_tokens = inputs.input_ids[0, answer_start_index : answer_end_index + 1]
23 print(tokenizer.decode(predict_answer_tokens, skip_special_tokens=True)) # print " nice puppet" or "Elephant" for the 2nd question and text
24
25 print(answer_start_index) # tensor(14)
26 print(answer_end_index) # tensor(15)

```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL

zsh + [icon] [icon] ... ^ x

```
thuangs38@ipsec-10-2-96-188 transformers_code % conda activate hugging-transformers
```

- (hugging-transformers) thuangs38@ipsec-10-2-96-188 transformers_code % python bart_question_answer.py
You passed along 'num_labels=3' with an incompatible id to label map: {'0': 'LABEL_0', '1': 'LABEL_1'}. The number of labels will be overwritten to 2.
You passed along 'num_labels=3' with an incompatible id to label map: {'0': 'LABEL_0', '1': 'LABEL_1'}. The number of labels will be overwritten to 2.
You passed along 'num_labels=3' with an incompatible id to label map: {'0': 'LABEL_0', '1': 'LABEL_1'}. The number of labels will be overwritten to 2.
You passed along 'num_labels=3' with an incompatible id to label map: {'0': 'LABEL_0', '1': 'LABEL_1'}. The number of labels will be overwritten to 2.

```
nice puppet
tensor(14)
tensor(15)
```

- (hugging-transformers) thuangs38@ipsec-10-2-96-188 transformers_code % python bart_zero_shot_classify.py
Downloading model.safetensors: 100% ██ | 1.63G/1.63G [00:59<00:00, 27.3MB/s]
{'sequence': 'one day I will see the world', 'labels': ['travel', 'dancing', 'cooking'], 'scores': [0.9938650727272034, 0.003273803973570466, 0.002861034125089645]}

Other Potential Models

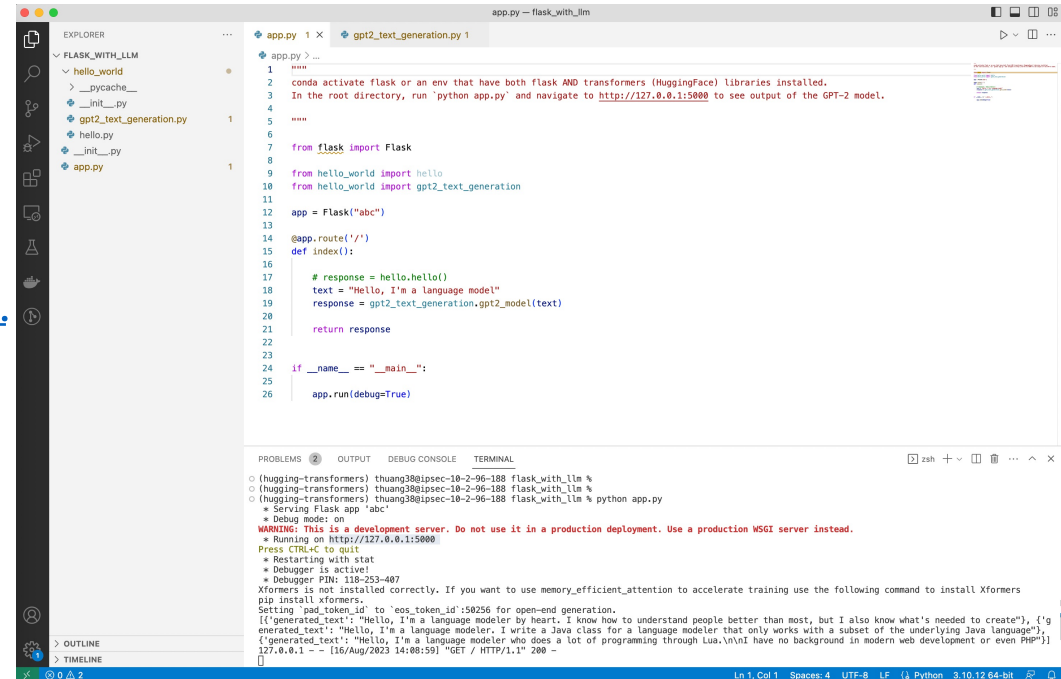
- Sentiment analysis:
 - Sentiment analysis task:
https://huggingface.co/models?pipeline_tag=text-classification&sort=downloads
- Group similar contents or posts:
 - Zero shot classification task:
https://huggingface.co/models?pipeline_tag=zero-shot-classification&sort=downloads
 - Sentence similarity task:
https://huggingface.co/models?pipeline_tag=sentence-similarity&sort=downloads
- Multiple models for each task available
- Example scripts available

Flask with Large Language Model

Flask: A web framework in Python allowing development of web applications

<https://flask.palletsprojects.com/en/2.3.x/installation/>

Example code on Canvas:
flask_with_llm



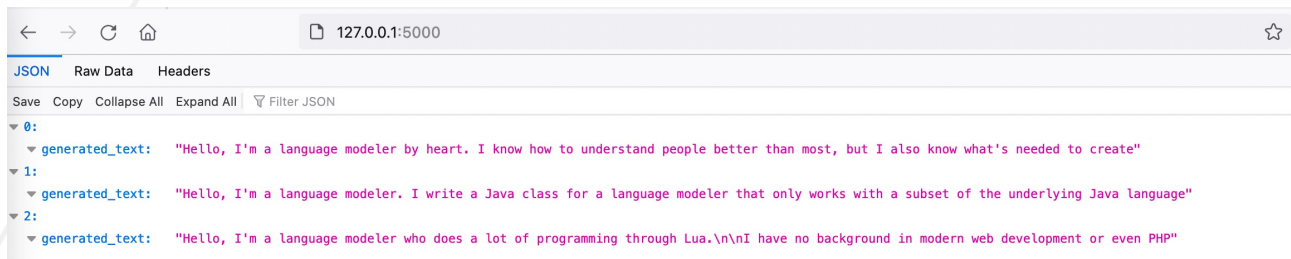
```
EXPLORER
  FLASK_WITH_LLM
    __pycache__
    __init__.py
    gpt2_text_generation.py
    hello.py
    __init__.py
    app.py

app.py
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gpt2_text_generation.py
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PROBLEMS
2
OUTPUT
DEBUG CONSOLE
TERMINAL

(hugging-transformers) thuang38@ipsec-10-2-96-188 flask_with_llm %
(hugging-transformers) thuang38@ipsec-10-2-96-188 flask_with_llm %
(hugging-transformers) thuang38@ipsec-10-2-96-188 flask_with_llm % python app.py
 * Serving Flask app 'abc'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 118-253-487
Xformers is not installed correctly. If you want to use memory_efficient_attention to accelerate training use the following command to install Xformers
pip install xformers.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
[{"generated_text": "Hello, I'm a language modeler by heart. I know how to understand people better than most, but I also know what's needed to create"}, {"generated_text": "Hello, I'm a language modeler. I write a Java class for a language modeler that only works with a subset of the underlying Java language"}, {"generated_text": "Hello, I'm a language modeler who does a lot of programming through Lua.\n\nI have no background in modern web development or even PHP"}]]
127.0.0.1 - - [16/Aug/2023 14:08:59] "GET / HTTP/1.1" 200 -
```



References

1. Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
2. Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. [BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880, Online. Association for Computational Linguistics.
3. Radford, A., Wu, J., Child, R., Luan, D., Amodei, D. & Sutskever, I. (2018). Language Models are Unsupervised Multitask Learners. , .